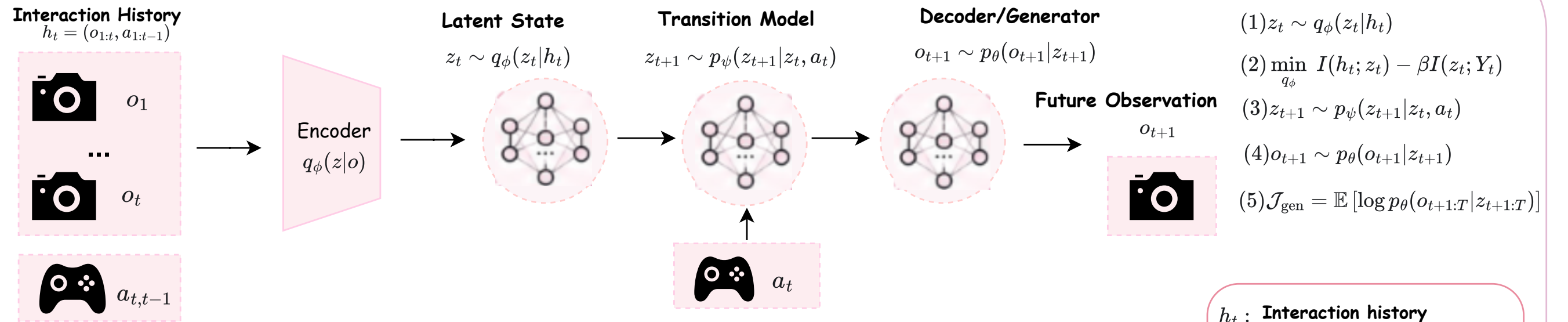


II.A. Representation and Prediction



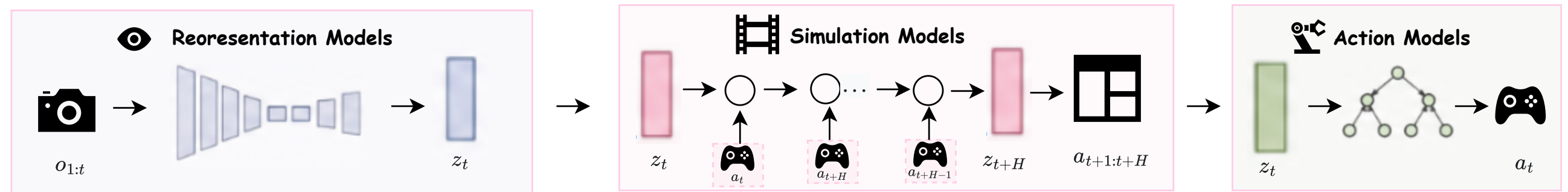
h_t : Interaction history
 z_t : Predictive latent state
 a_t : Action
 o_t : Observation
 Y_t : Future-dependent variables

II.B. Compression-Generation Principle

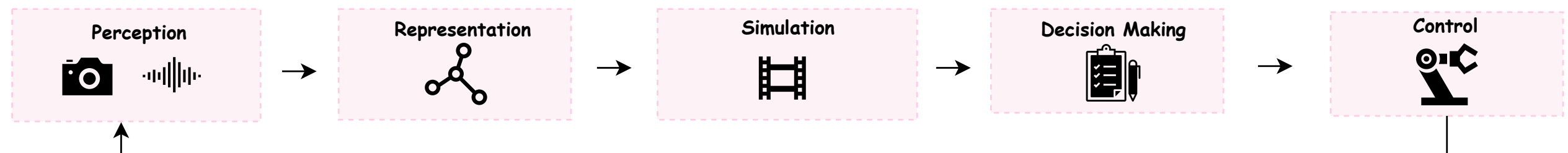
$$\max_{q_\phi} I(z_t; Y_t) - \lambda I(h_t; z_t), \quad z_t \sim q_\phi(z_t|h_t) \quad (6)$$

$$\longleftrightarrow \min_{q_\phi} I(h_t; z_t) \quad \text{s.t.} \quad I(z_t; Y_t) \geq \eta \quad (7)$$

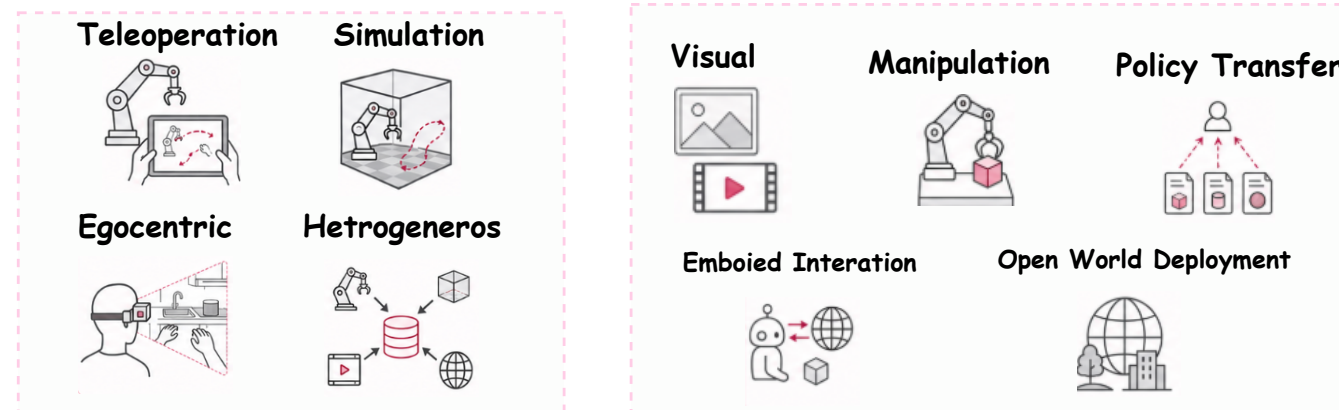
III. Information-Flow taxonomy



IV. Embodied World Models



V. Datasets & Benchmarks



VI. Open Challenges

